

Assignment - 01

Section - A

1. What are the features of data warehousing?

- Ans. i) Subject Oriented
 ii) Integrated
 iii) Non-volatile
 iv) Time-variant.

2. What is the diff. b/w DM and DW?

Ans. Data Warehousing

- Stores & manages historical data.
- Focuses on data storage, integration & retrieval.
- Deals with structured, consistent data.
- Used mainly by IT professionals for data management.

Data Mining.

- Analyzes data to extract patterns & knowledge.
- Focuses on decision making & prediction.
- Can work on structured and unstructured data.
- Used by business analysts to support strategic decisions.

3. What are the benefits of data mining?

- Ans. i) Helps in discovering hidden patterns.
 ii) Predicts future trends and behaviours.

iii) Improves decision-making.

4. What is Data-mining?

Ans: It is a process of discovering hidden patterns, relationships, knowledge from large data sets. It involves techniques like classification, clustering, regression and association.

5. What type of info is collected in Data-mining?

Ans: i) Patterns & Trends

ii) Associations / Relationships

iii) Classification

iv) Clustering

v) Prediction

vi) Correlations & Exceptions

Section - B

1. Discuss all steps of data warehouse process.

Ans: i) Data Extraction: It is a process of collecting data for the warehouse from various sources. Data may come from: Production data, legacy data, Meta data, Other systems.

- ii) **Data Cleaning**: Detects errors in data & rectifies them when possible. Techniques include:
- Using transformation rules
 - Using domain & integrity knowledge.
- iii) **Data Transformation**: Converts data from legacy or heterogeneous formats into warehouse format. It ensures consistency & standardization of data.
- iv) **Data Loading**: Loads the processed data into the warehouse by sorting, summarizing, verifying, checking integrity & partitioning.
- v) **Data Refreshing**: Propagates the updates from data sources to the warehouse. Keeps the data warehouse upto date with source system.

2. What kinds of data can be mined?

Ans: Data can be mined from flat files, relational databases, DWs, transaction databases, multimedia, spatial databases and the WWW.

- i) **Flat Files**: Simple data files in text or binary format. It can store transaction data, scientific

measurements, or time-series data.

- ii) **Relational databases:** Consist of a set of tables. Databases can analyze relationships in relational DBs.
- iii) **Data Warehouses:** Repositories of data collected from multiple sources. Usually modeled by a multi-dimensional structure to support decision-making.
- iv) **Transaction Databases:** Contain a set of records representing transactions with a time stamp.
- v) **Multimedia Databases:** Contain data such as images, videos, audio, and text media.
- vi) **Spatial Databases:** Store geographical or spatial data like maps.

3. Discuss functionality of DM.

Ans: i) **Concept / Class description:**

- Summarization of general characteristics of data.
- Includes characterization and discrimination.

ii) **Association Analysis:**

- Finds relationships & co-relations among items in large datasets. Eg: Market Basket analysis.

ii) Classification:

- Assigns data to predefined categories or classes using models like decision trees, neural n/ws etc.
Eg: Classifying emails as spam or not spam.

iii) Prediction:

- Forecasts future data values or trends based on current & historical data. Eg: predicting customer churn or stock prices.

iv) Clustering

- Groups similar obj into clusters without predefined labels. Eg: Customer segmentation in marketing.

Section - C

1. Define KDD. Identify & describes phases in KDD process.

Ans Knowledge Discovery in Database is the overall process of extracting valid, novel, potentially useful & understandable patterns or knowledge from large

volumes of data.

Phases of KDD process:

i) Data Selection:

- Identify the data relevant to the analysis task.
- Data is collected from multiple sources such as databases, data warehouses, flat files or online repositories.

ii) Data Preprocessing (Cleaning):

- Detect and remove noise, inconsistencies, duplication and missing values.
- Ensures the quality & reliability of data. Eg: correcting errors in customers records.

iii) Data Transformation:

- Transform data into suitable forms of mining.
- Includes normalization, aggregation, feature extraction and dimensionality reduction.

iv) Data Mining:

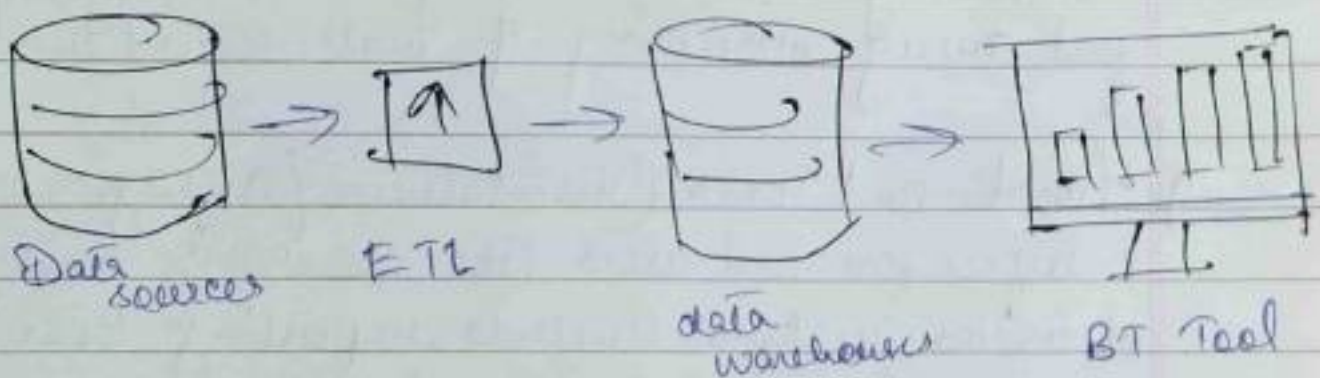
- The core steps of KDD.
- Apply intelligent methods to extract patterns from data.

v) Pattern Evaluation:

- Evaluate mined patterns to identify which are valid, useful, novel and interesting.
- Removes irrelevant or redundant results.

2. Discuss Architecture of DW with diagram.

Ans: It is a framework that shows how data flows from operational system to the warehouse & finally to end-users for analysis and reporting.



It typically consists of following layers.

- Data Sources:** It comes from diff. sources like: ERP systems, legacy systems, External data sources.
- Data staging area (ETL-Extracting, Transformation, loading):** Data is extracted from sources cleaned and transformed into a consistent format.

- iii) Data Storage: The central repository that stores integrated-subject-oriented, time-variant & non-volatile data. Organised in two ways:
- Enterprise Warehouse
 - Data Marts

iv) Meta Data: Stores info about the data. It helps in managing & using the data warehouse efficiently.

v) OLAP Engine (Online Analytical Processing): Supports fast query processing & multidimensional analysis.

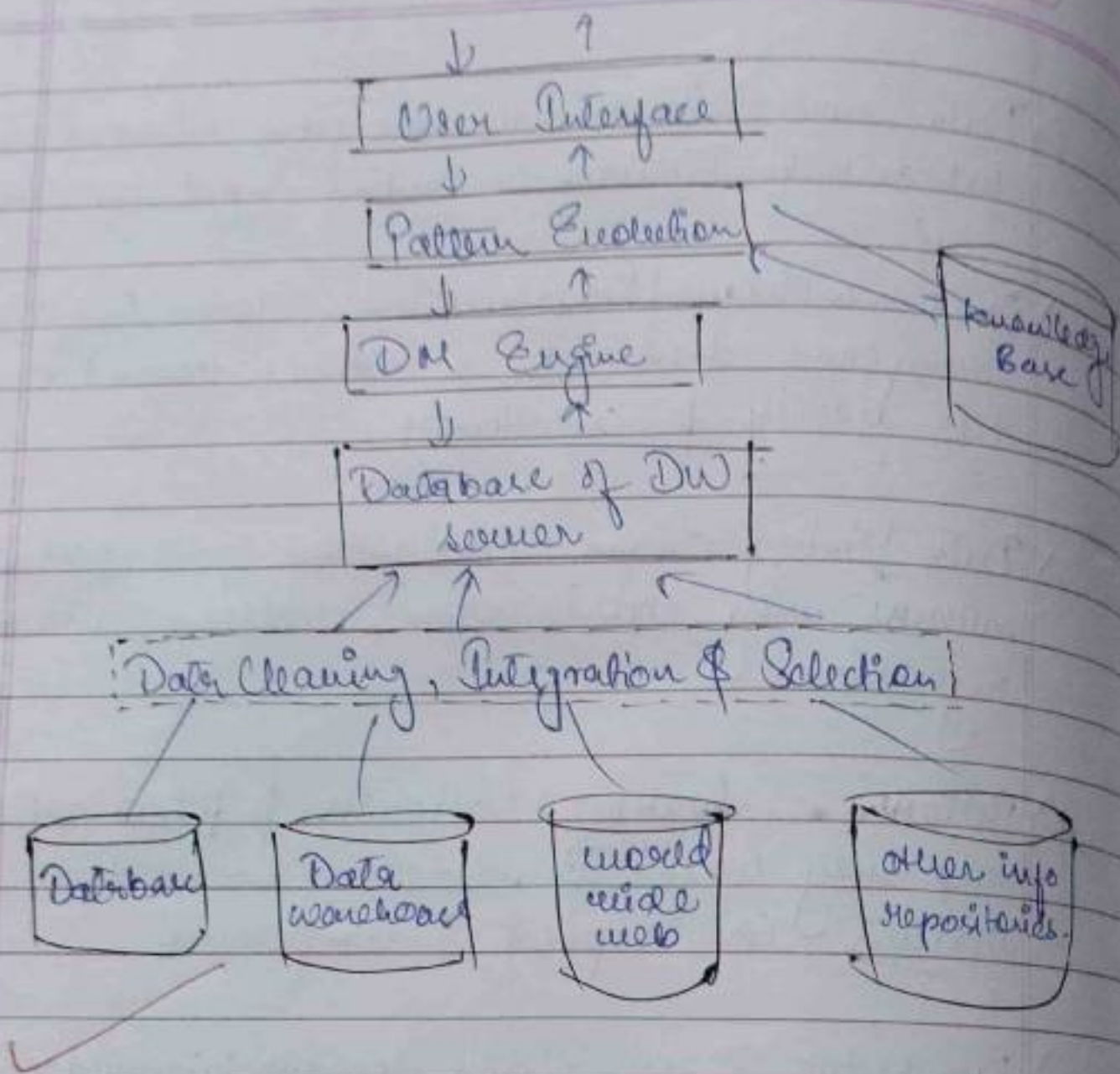
vi) Front-End Tools (Presentation / Analysis layer): Interface for end users like managers, analysts & decision makers. Include reporting tools, dashboards, visualization, DM, and business intelligence apps.

3. Discuss Architecture of DM systems.

Ans: It describes the components & their interactions to extract useful knowledge from large datasets. It integrates database, DWS, mining engines & user interfaces.

Components of DM systems Architecture:

- i) Data Source: Stores raw data from databases, warehouses, text or web. Provides the initial input for mining.
- ii) Data Warehouse / Preprocessing: Cleans, integrates & transforms data into a suitable format. Ensures data quality & consistency.
- iii) Data Mining Engine: Core module that discovers hidden patterns using classification, clustering, association & regression.
- iv) Patterns Evaluation: Validates & filters patterns to keep only useful & interesting knowledge. Applies measures like support & confidence.
- v) Knowledge Base: Stores domain knowledge, rules, constraints & meta data. Guides the mining process to improve pattern quality.
- vi) User Interface: Allows users to define mining tasks & interact with the systems. Visualizes results through charts, graphs, or reports.



Data Mining Architecture.

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